FAST- National University of Computer and Emerging Sciences

Multi-Variable Calculus

MT-1008

Assignment # 2

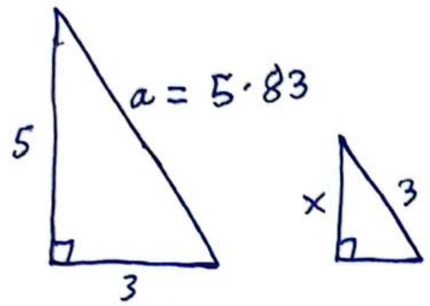
SUBMITTED TO: Mr. Imran Shahzad

SUBMITTED BY: Muhammad Moiz Ansari

SECTION: BS(CS) - F

Roll No: 23i - 0523

Q-2:



$x \rightarrow$ Shortest distance

$\therefore x^2 = (5)^2 + (3)^2$

$x = \sqrt{25+9}$

$x = \sqrt{34}$

$x = 5.83$ mi

By similar triangles,

$$\frac{x}{5} = \frac{3}{5.83}$$

$$x = \frac{3}{5.83}(5)$$

$\boxed{x = 2.57 \text{ mi}}$ Ans. → Shortest distance

Cost $= (x)\cdot(250{,}000)$

Cost $= (2.57)(250{,}000)$

$\boxed{\text{Cost} = \$642\ 500}$ Ans.

②

Q-3:

$f(x,y) = \sin \pi x \cos \pi y \qquad R \to 0 \le x \le \frac{1}{4}, \ \frac{1}{4} \le y \le \frac{1}{2}$

(a)

Python surface attached at end

(b)

$0 \le \iint_R f(x,y)\, dA \le \frac{1}{32} \to$ To show

$$\Rightarrow \int_{1/4}^{1/2} \int_{0}^{1/4} \sin \pi x \cos \pi y \, dx\, dy$$

$$= \int_{1/4}^{1/2} \cos \pi y \int_{0}^{1/4} \sin \pi x \, dx\, dy$$

$$= \int_{1/4}^{1/2} \cos \pi y \left[\frac{-\cos \pi x}{\pi} \right]_0^{1/4} dy$$

$$= \int_{1/4}^{1/2} \cos \pi y \left(-\frac{\cos\left(\pi \times \frac{1}{4}\right)}{\pi} - \left(-\frac{\cos \pi (0)}{\pi} \right) \right) dy$$

$$= \int_{1/4}^{1/2} \cos \pi y \left(-\frac{\sqrt{2}}{2\pi} + \frac{1}{\pi} \right) dy$$

$$= \left(\frac{1}{\pi} - \frac{\sqrt{2}}{2\pi} \right) \int_{1/4}^{1/2} \cos \pi y \, dy$$

$$= (0.093) \left[\frac{\sin \pi y}{\pi} \right]_{1/4}^{1/2}$$

$$= (0.093)\left[\frac{\sin \pi\left(\frac{1}{2}\right)}{\pi} - \frac{\sin \pi\left(\frac{1}{4}\right)}{\pi}\right]$$

$$= (0.093)\left(\frac{1}{\pi} - \frac{\sqrt{2}}{2\pi}\right)$$

$$= (0.093)(0.093)$$

$$= 0.0087$$

$\Rightarrow 0 \le 0.0087 \le \frac{1}{32}$ → Hence proved.

Q-4:

$V = ?$, $x^2 + 2y^2 + z = 16$, $x = 2, y = 2, x = 0, y = 0, z = 0$

$\Rightarrow \boxed{z = 16 - x^2 - 2y^2}$

$R \rightarrow 0 \le x \le 2, 0 \le y \le 2$

$$\Rightarrow V = \iint_R z \, dA$$

$$= \int_0^2 \int_0^2 (16 - x^2 - 2y^2)\, dx\, dy$$

$$= \int_0^2 \left[16x - \frac{x^3}{3} - 2y^2x\right]_0^2 dy$$

$$= \int_0^2 \left(32 - \frac{8}{3} - 4y^2\right) - 0 \, dy$$

$$= \left[32y - \frac{8y}{3} - \frac{4y^3}{3}\right]_0^2$$

$$= 64 - \frac{16}{3} - \frac{4(8)}{3} = 48$$

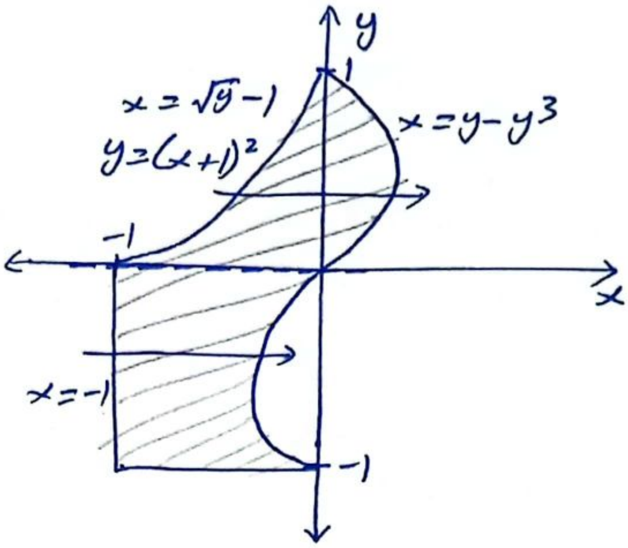
$\Rightarrow \boxed{\text{Volume} = 48 \text{ cubic units}}$ Ans.

④

Q-6:

We divide the region from x-axis

$$\text{Area} = \int_{-1}^{0}\int_{-1}^{y-y^3} (1)\,dx\,dy + \int_{0}^{1}\int_{\sqrt{y}-1}^{y-y^3} (1)\,dx\,dy$$



$$= \int_{-1}^{0} [x]_{-1}^{y-y^3}\,dy + \int_{0}^{1} [x]_{\sqrt{y}-1}^{y-y^3}\,dy$$

$$= \int_{-1}^{0} [y-y^3-(-1)]\,dy + \int_{0}^{1} [y-y^3-(\sqrt{y}-1)]\,dy$$

$$\left[\begin{array}{l} y=(x+1)^2 \\ \Rightarrow \sqrt{y} = \sqrt{(x+1)^{\cancel{2}}} \\ \sqrt{y}-1 = x \\ \underline{x=\sqrt{y}-1} \end{array}\right.$$

$$= \int_{-1}^{0} (y-y^3+1)\,dy + \int_{0}^{1} (y-y^3-\sqrt{y}+1)\,dy$$

$$= \left[\frac{y^2}{2} - \frac{y^4}{4} + y\right]_{-1}^{0} + \left[\frac{y^2}{2} - \frac{y^4}{4} - \frac{y^{3/2}}{\frac{3}{2}} + y\right]_{0}^{1}$$

$$= (0) - \left(\frac{1}{2} - \frac{1}{4} - 1\right) + \frac{1}{2} - \frac{1}{4} - \frac{2}{3} + 1 - (0)$$

$$= -\cancel{\frac{1}{2}} + \cancel{\frac{1}{4}} + 1 + \cancel{\frac{1}{2}} - \cancel{\frac{1}{4}} - \frac{2}{3} + 1$$

$\boxed{\text{Area} = \frac{4}{3} \text{ sq. units}}$ Ans.

Q-9:

(a)

$z = f(x, y) = x^2 + 3xy - y^2$

$dz = ?$

$\Rightarrow dz = 2x\,dx + 3(x\,dy + y\,dx) - 2y\,dy$

$dz = 2x\,dx + 3x\,dy + 3y\,dx - 2y\,dy$

$\boxed{dz = (2x + 3y)dx + (3x - 2y)dy}$ Ans. — 1

(b)

$dx = 2.05 - 2$ | $dy = 2.96 - 3$

$dx = 0.05$ | $dy = -0.04$

$\Delta z = z_{final} - z_{initial}$ — 2

$z_{initial} = (2)^2 + 3(2)(3) - (3)^2$

$z_{initial} = 13$

$z_{final} = (2.05)^2 + 3(2.05)(2.96) - (2.96)^2$

$z_{final} = 13.6449$

2 $\Rightarrow \Delta z = 13.6449 - 13$

$\boxed{\Delta z = 0.6449}$

1 $\Rightarrow dz = [2(2) + 3(3)](0.05) + [3(2) - 2(3)](-0.04)$

$\boxed{dz = 0.65}$

Q-1:

$z = x^2 + 4y^2$ — ①, $x^2 + y^2 = 1 \rightarrow x^2 + y^2 - 1 = 0$ — ②

$\Rightarrow f(x, y) = x^2 + 4y^2$ $\qquad$ $g(x, y) = x^2 + y^2 - 1$

$f_x = 2x$, $f_y = 8y$, $g_x = 2x$, $g_y = 2y$

$f_x = \lambda g_x$

$\cancel{2}x = \lambda \cancel{2}x$

$x - \lambda x = 0$

$x(1 - \lambda) = 0$

$x = 0$ or $1 - \lambda = 0$

$\lambda = 1$

$f_y = \lambda g_y$

$\overset{4}{\cancel{8}}y = \lambda \overset{1}{\cancel{2}}y$

$4y - \lambda y = 0$

$y(4 - \lambda) = 0$

$y = 0$ or $4 - \lambda = 0$

$\lambda = 4$

When $x = 0$,

② $\Rightarrow (0)^2 + y^2 - 1 = 0$

$y = 1$

① $\Rightarrow z = (0)^2 + 4(1)^2$

$z = 4$

$f(x, y) = 4$ at $(0, 1)$

↳ Maxima

When $y = 0$,

② $\Rightarrow x^2 + (0)^2 - 1 = 0$

$x = 1$

① $\Rightarrow z = (1)^2 + 4(0)^2$

$z = 1$

$f(x, y) = 1$ at $(1, 0)$

↳ Minima

$\boxed{\text{Highest point on curve} = (0, 1, 4)}$ Ans.

$\boxed{\text{Lowest point on curve} = (1, 0, 1)}$ Ans.

Q-10:

$$f(x,y) = \frac{10000\, e^{y}}{1+\frac{|x|}{2}} \quad , -5 \le x \le 5 \, , -2 \le y \le 0$$

$$\int_{-5}^{5}\int_{-2}^{0} f(x,y)\, dy\, dx = ?$$

$$= \int_{-5}^{5}\int_{-2}^{0} \left(\frac{10000\, e^{y}}{1+\frac{|x|}{2}}\right) dy\, dx$$

$$= \int_{-5}^{5} \frac{10000}{1+\frac{|x|}{2}} \left[e^{y}\right]_{-2}^{0} dx$$

$$= \int_{-5}^{5} \frac{10000}{1+\frac{|x|}{2}} \left(e^{0}-e^{-2}\right) dx$$

$$= 10000\left(1-e^{-2}\right) \int_{-5}^{5} \frac{1}{1+\frac{|x|}{2}}\, dx$$

$$= 10000\left(1-e^{-2}\right) \left[\int_{-5}^{0} \frac{1}{1-\frac{x}{2}}\, dx + \int_{0}^{5} \frac{1}{1+\frac{x}{2}}\, dx\right]$$

$$= 8646.65 \left[\left[\frac{\ln\left(1-\frac{x}{2}\right)}{-\frac{1}{2}}\right]_{-5}^{0} + \left[\frac{\ln\left(1+\frac{x}{2}\right)}{\frac{1}{2}}\right]_{0}^{5}\right]$$

⑧

$$= 8646.65\left[-2\left[\cancel{\ln(1)}^{0} - \ln\left(1-\frac{(5)}{2}\right)\right] + 2\left[\ln\left(1+\frac{5}{2}\right) - \cancel{\ln 1}^{0}\right]\right]$$

$$= 8646.65\left(2\ln\left(1+\frac{5}{2}\right) + 2\ln\left(1+\frac{5}{2}\right)\right)$$

$$= 8646.65\left(4\ln\left(\frac{7}{2}\right)\right)$$

$\boxed{= 43328.81 \text{ bacteria}}$ Ans

Q-7:

$z = x^2+y^2$, $3x^2+2y^2+z^2 = 9$, $P_0(1,1,2)$

$z - x^2 - y^2 = 0$, $3x^2+2y^2+z^2-9=0$

$f(x,y,z) = z - x^2 - y^2$, $g(x,y,z) = 3x^2+2y^2+z^2-9$

$$\nabla f = f_x\hat{i} + f_y\hat{j} + f_z\hat{k}$$

$$= -2x\hat{i} + (-2y)\hat{j} + \hat{k}$$

$$= -2x\hat{i} - 2y\hat{j} + \hat{k}$$

$$\nabla f_{(1,1,2)} = -2\hat{i} - 2\hat{j} + \hat{k}$$

$$\nabla g = 6x\hat{i} + 4y\hat{j} + 2z\hat{k}$$

$$\nabla g_{(1,1,2)} = 6\hat{i} + 4\hat{j} + 4\hat{k}$$

$$\vec{V} = \nabla f \times \nabla g$$

$$= \begin{vmatrix} i & j & k \\ -2 & -2 & 1 \\ 6 & 4 & 4 \end{vmatrix}$$

$$= \hat{i}(-8-4) - \hat{j}(-8-6) + \hat{k}(-8+12)$$

$\vec{v} = -12\hat{i} + 14\hat{j} + 4\hat{k}$

$\overrightarrow{PP_0} = (x-x_0)\hat{i} + (y-y_0)\hat{j} + (z-z_0)\hat{k}$

$= (x-1)\hat{i} + (y-1)\hat{j} + (z-2)\hat{k}$

$\overrightarrow{PP_0} = \vec{v}\,t$

$x-1=-12t$, $y-1=14t$, $z-2=4t$

$\boxed{x=-12t+1}$, $\boxed{y=14t+1}$, $\boxed{z=4t+2}$

Q-8: (b)

(a) Python surface attached at end.

$f(x,y) = x e^{xy}$ at $P_0(1,0)$

$f_x = x e^{xy} y + e^{xy}$

$f(x_0,y_0) = 1 e^{1(0)}$

$= 1$

$f_x(x_0,y_0) = (1) e^{1(0)}(0) + e^{1(0)}$

$= 1$

$f_y = x e^{xy} x$

$= x^2 e^{xy}$

$f_y(x_0,y_0) = (1)^2 e^{1(0)}$

$= 1$

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

$= 1 + 1(x-1) + 1(y-0)$

$= \cancel{1} + x - \cancel{1} + y$

$\boxed{L(x,y) = x + y}$

$$\therefore f(x_0, y_0) \approx L(x_0, y_0) \text{ —— } ①$$

$$L(1.1, -0.1) = 1.1 - 0.1$$
$$= 1 \text{ —— } ②$$

$$① \,\&\, ② \Rightarrow \boxed{f(x_0, y_0) \approx 1} \quad \underline{\text{Ans.}}$$

Q-5:

$V = ?$ $\quad x + 2y + z = 2$ —— ①, $\quad x = 2y$ —— ②, $\quad x = 0, \quad z = 0$

$z = 0$ in eq. ①:

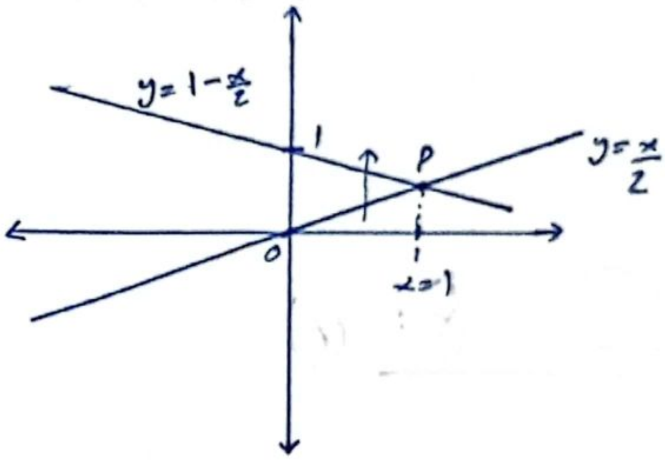
$$x + 2y = 2$$
$$2y = 2 - x$$
$$y = \frac{2 - x}{2}$$
$$y = 1 - \frac{x}{2}$$

② $\Rightarrow$ $y = \frac{x}{2}$

At P;

$y = \frac{x}{2}$ in $y = 1 - \frac{x}{2}$:

$$\frac{x}{2} = 1 - \frac{x}{2}$$
$$\frac{\cancel{2}x}{\cancel{2}} = 1$$
$$x = 1$$



$$① \Rightarrow z = 2 - x - 2y$$

$$\Rightarrow V = \int_0^1 \int_{\frac{x}{2}}^{1-\frac{x}{2}} 2 - x - 2y \; dy \, dx$$

$$= \int_0^1 \left[2y - xy - \frac{\cancel{2}y^2}{\cancel{2}}\right]_{\frac{x}{2}}^{1-\frac{x}{2}} dx$$

$$= \int_0^1 2\left(1 - \frac{x}{2}\right) - x\left(1 - \frac{x}{2}\right) - \left(1 - \frac{x}{2}\right)^2 - \left[\cancel{2}\left(\frac{x}{\cancel{2}}\right) - \frac{x^2}{2} - \frac{x^2}{4}\right] dx$$

$$= \int_0^1 2 - \frac{\cancel{2}x}{\cancel{2}} - x + \frac{x^2}{2} - \left(1 + \frac{x^2}{4} - \frac{\cancel{2}x}{\cancel{2}}\right) - x + \frac{x^2}{2} + \frac{x^2}{4} \; dx$$

$$= \int_0^1 2 - 3x + x^2 + \cancel{\frac{x^2}{4}} - 1 - \cancel{\frac{x^2}{4}} + x \, dx$$

$$= \int_0^1 1 - 2x + x^2 \, dx$$

$$= \left[x - \frac{\cancel{2}x^2}{\cancel{2}} + \frac{x^3}{3} \right]_0^1$$

$$= \cancel{1} - \cancel{1} + \frac{1}{3} - (0)$$

$$\boxed{V = \frac{1}{3} \text{ cubic units}} \text{ Ans.}$$

Q-11:

$x^2 + y^2 = r^2 \Rightarrow x^2 + y^2 = (20)^2$ —(1) Depth = $2 \rightarrow 7$

$\because d = 40$

$r = \frac{d}{2}$

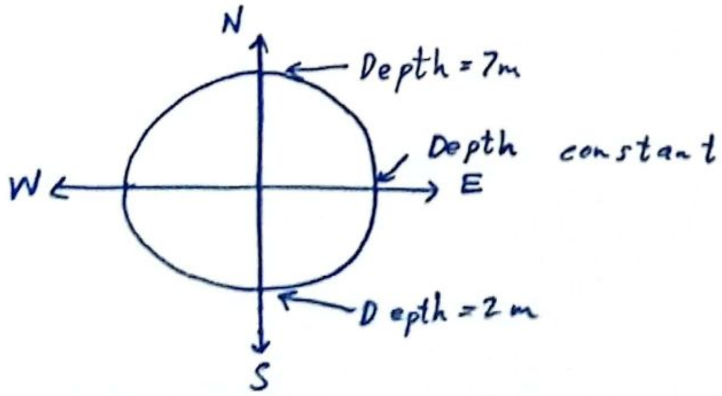
$r = \frac{40}{2}$

$r = 20$

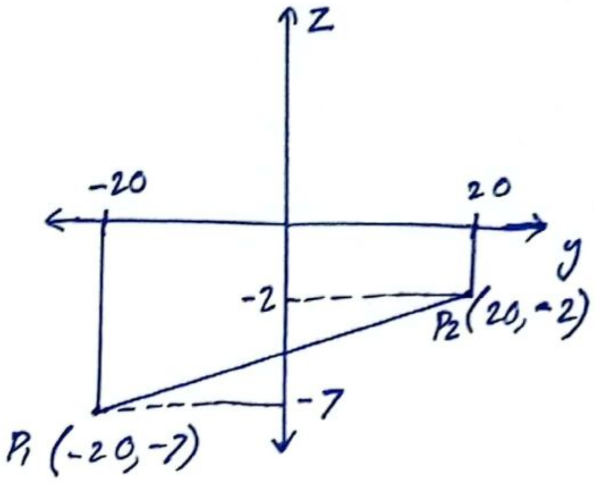
$\Rightarrow x^2 + y^2 = 400$

$V = ?$

x is constant when calculating depth of pool, since depth is constant along east and west lines.



(12)



$$\text{Slope} = m = \frac{z_2 - z_1}{y_2 - y_1}$$

$$= \frac{-2-(-7)}{20-(-20)}$$

$$= \frac{-2+7}{20+20}$$

$$= \frac{5}{40}$$

$$\boxed{m = \frac{1}{8}}$$

$$\because z - z_1 = m(y - y_1)$$

$$\Rightarrow z + 7 = \frac{1}{8}(y+20)$$

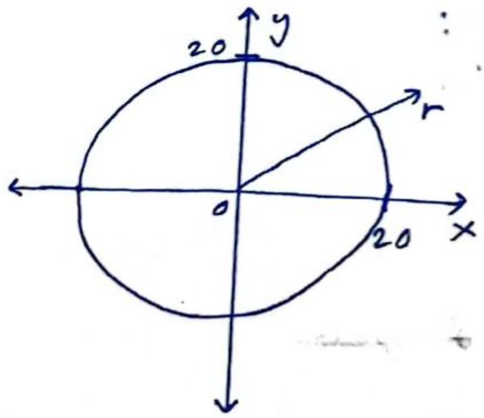
$$z = \frac{y}{8} + \frac{20}{8} - 7$$

$$\boxed{z = \frac{y}{8} - \frac{9}{2}} \text{ --- (2)}$$

Now, for limits:

$r : 0 \rightarrow 20$

$\theta : 0 \rightarrow 2\pi$



$$\Rightarrow V = \int_0^{2\pi}\int_0^{20} z\,r\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{20}\left(\frac{r\sin\theta}{8} - \frac{9}{2}\right) r\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{20} \frac{r^2\sin\theta}{8} - \frac{9r}{2}\,dr\,d\theta$$

$$= \int_0^{2\pi}\left[\frac{r^3}{3}\frac{\sin\theta}{8} - \frac{9r^2}{4}\right]_0^{20} d\theta$$

$$= \int_0^{2\pi} \frac{(20)^3}{3} \times \frac{\sin\theta}{8} - \frac{9(20)^2}{4} - (0)\, d\theta$$

$$= \int_0^{2\pi} \frac{1000}{3} \sin\theta - 900\, d\theta$$

$$= \left[\frac{1000}{3}(-\cos\theta) - 900\theta \right]_0^{2\pi}$$

$$= \frac{-1000}{3}\cos(2\pi) - 900(2\pi) - \left[\frac{-1000}{3}\cos(0) - 900(0) \right]$$

$$= -\cancel{\frac{1000}{3}} - 1800\pi + \cancel{\frac{1000}{3}} + 0$$

$$V = -1800\pi$$

$-ve$ sign indicates volume is under y-axis (below $z=0$)

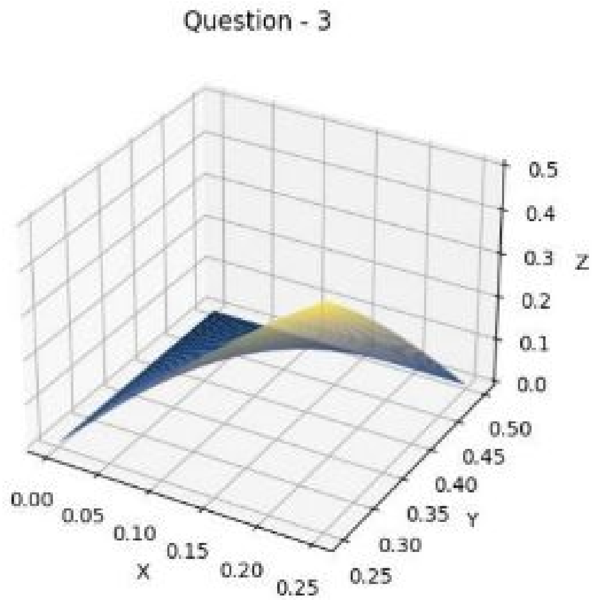
$\Rightarrow \boxed{V = 1800\pi \ m^3}$ Ans.

Question-3

Code:

```
#------ Importing libraries ------
import numpy as np
import matplotlib.pyplot as plt
import math
from mpl_toolkits.mplot3d import Axes3D

#------ Defining function ------
def E(x, y, z):
    return np.sin(np.pi * x) * np.cos(np.pi * y)  # Use numpy functions for
element-wise operations

#------ x & y values ------
x = np.linspace(0, 0.25, 100)
y = np.linspace(0.25, 0.5, 100)
X, Y = np.meshgrid(x, y)  # Generate meshgrid for X and Y

Z = E(X, Y, 0)  # Calculate Z values using the function E

#------ Plotting surface ------
figure = plt.figure()
graph = figure.add_subplot(111, projection='3d')
graph.plot_surface(X, Y, Z, cmap='cividis')

#------ Labelling ------
graph.set_xlabel('X')
graph.set_ylabel('Y')
graph.set_zlabel('Z')
graph.set_title('Question - 3')

plt.show()
```

Graph:



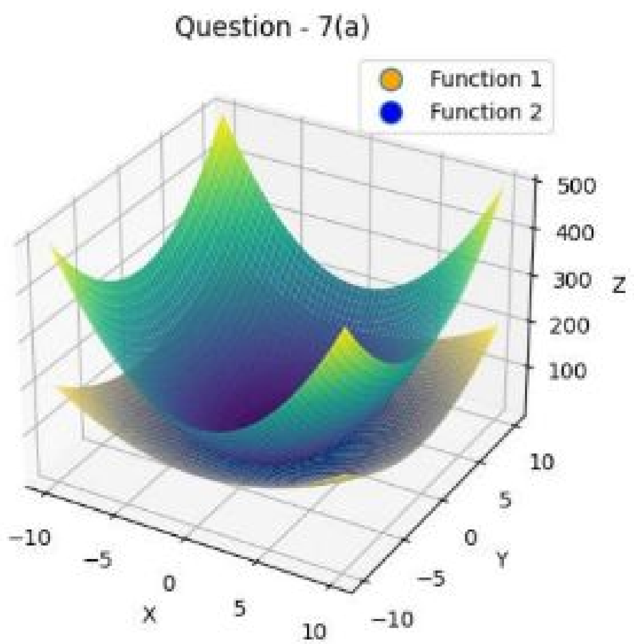
Question-7(a)

Code:

```
#------ Importing libraries ------
import numpy as np
import matplotlib.pyplot as plt
import math
from mpl_toolkits.mplot3d import Axes3D

#------ Defining functions ------
def E1(x, y, z):
    return pow(x, 2) + pow(y, 2)

def E2(x, y, z):
    return 3 * pow(x, 2) + 2 * pow(y, 2) + pow(z, 2)

#------ x & y values ------
x = np.linspace(-10, 10, 100)
y = np.linspace(-10, 10, 100)
X, Y = np.meshgrid(x, y)

#------ Calculate Z values for both functions ------
Z1 = E1(X, Y, 0)
```

```
Z2 = E2(X, Y, 0)

#------ Plotting surface ------
figure = plt.figure()
graph = figure.add_subplot(111, projection='3d')

#------ Plotting surface for first function ------
surf1 = graph.plot_surface(X, Y, Z1, cmap='cividis')

#------ Plotting surface for second function ------
surf2 = graph.plot_surface(X, Y, Z2, cmap='viridis')

#------ Labelling ------
graph.set_xlabel('X')
graph.set_ylabel('Y')
graph.set_zlabel('Z')
graph.set_title('Question - 7(a)')

#------ Creating a 2D plot for the legend ------
d1 = plt.Line2D([0], [0], linestyle="none", marker='o', markersize=10,
markerfacecolor='orange')
d2 = plt.Line2D([0], [0], linestyle="none", marker='o', markersize=10,
markerfacecolor='blue')
plt.legend([d1, d2], ['Function 1', 'Function 2'])

plt.show()
```

Graph:

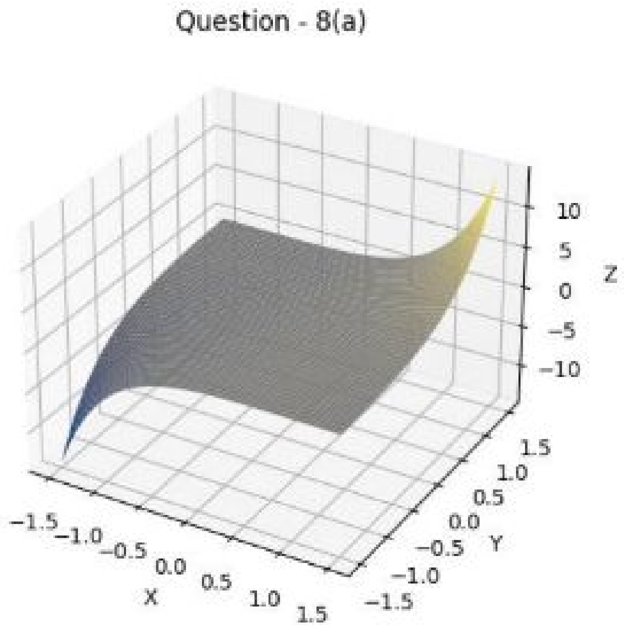


Question-8(a)

Code:

```
#------ Importing libraries ------
import numpy as np
import matplotlib.pyplot as plt
import math
from mpl_toolkits.mplot3d import Axes3D

#------ Defining function ------
def E(x, y, z):
    return x*pow(math.e, x*y)

#------ x & y values ------
x = np.linspace(-1.5, 1.5, 100)
y = np.linspace(-1.5, 1.5, 100)
X, Y = np.meshgrid(x, y)

Z = E(X, Y, 0)

#------ Plotting surface ------
figure = plt.figure()
graph = figure.add_subplot(111, projection='3d')
graph.plot_surface(X, Y, Z, cmap='cividis')

#------ Labelling ------
graph.set_xlabel('X')
graph.set_ylabel('Y')
graph.set_zlabel('Z')
graph.set_title('Question - 8(a)')

plt.show()
```

Graph:



Question-10

Code(Solution):

```
#from sympy import symbols, integrate, exp, Abs

x, y = symbols('x y')

# Function definition
expr = (10000 * exp(y)) / (1 + Abs(x) / 2)

# x limits:
a = -5
b = 5

# y limits:
c = -2
d = 0

# Integration:
in_1 = integrate(expr, (x, a, b))  # integrate wrt x
in_2 = integrate(in_1, (y, c, d))  # integrate wrt y
```

```
# Simplifying result
result = in_2.evalf()

# Printing
print("Result =", result)
```

Output:

```
Result = 43328.7974930283
```

Code(Graph):

```
#------ Importing libraries ------
import numpy as np
import matplotlib.pyplot as plt
import math
from mpl_toolkits.mplot3d import Axes3D

#------ Defining function ------
def E(x, y, z):
    return (10000*pow(math.e,y)) / (1+abs(x)/2)

#------ x & y values ------
x = np.linspace(-5, 5, 100)
y = np.linspace(-2, 0, 100)
X, Y = np.meshgrid(x, y)

Z = E(X, Y, 0)

#------ Plotting surface ------
figure = plt.figure()
graph = figure.add_subplot(111, projection='3d')
graph.plot_surface(X, Y, Z, cmap='cividis')

#------ Labelling ------
graph.set_xlabel('X')
graph.set_ylabel('Y')
graph.set_zlabel('Z')
graph.set_title('Question - 10')

plt.show()
```

Graph: